

Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

Project Design Phase-II: Data Flow Diagram & User Stories

Date:	October 3, 2026	Maximum Marks:	4 Marks
Team ID:	SWTID-2026-2585	Project Name:	Standard Laptop Automation

1. Data Flow Diagrams (DFD)

A Data Flow Diagram (DFD) visually represents how information flows through the IT procurement and automated workflow system.

DFD Level 0 (Context Diagram)

- **External Entities:**
 - **Customer / Employee:** Submits the standard laptop request.
 - **Manager / Approver:** Approves or rejects the service request.
 - **Hardware Team:** Receives the auto-configured catalog task for laptop setup.
- **System Process: ServiceNow Automated Procurement Platform** (Processes orders, executes Flow Designer logic, and manages tasks).
- **Data Flows In:** Laptop order details, approval confirmation.
- **Data Flows Out:** Service request status, catalog task notifications, assignment updates.

DFD Level 1 (Expanded Workflow Process)

1. **Process 1.1 (Catalog Browsing & Ordering):** The Customer selects the "Standard Laptop" item from the ServiceNow Service Catalog and submits the request → Generates a Requested Item (RITM) record.
2. **Process 1.2 (Approval Gate):** The system routes the order to the designated Approver → Approver reviews and sets status to **Approved**.
3. **Process 1.3 (Flow Designer Automation):** Upon approval, the Flow Designer trigger fires automatically → System executes the "Standard Laptop Task" flow.

4. **Process 1.4 (Task Generation & Assignment):** The flow creates a Catalog Task, updates the short description to “*Laptop need to Configured*”, and assigns the task directly to the **Hardware** group.

2. User Stories Table

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Customer (Web User)	Service Catalog Ordering	USN-1	As an employee, I want to browse and order a standard laptop from the ServiceNow hardware catalog so that my procurement request is officially logged.	I can view the standard laptop item and successfully click "Order Now"	High	Sprint-1
Customer (Web User)	Request Tracking	USN-2	As an employee, I want to track the status of my laptop request and view its approval state so that I know when it is progressing	I can open my request number and view real-time status updates.	High	Sprint-1
Approver / Manager	Request Approval	USN-3	As a manager, I want to receive and approve standard laptop requests quickly so that fulfillment can begin without delay.	I can access the approver section, right-click, and select "Approve" to trigger downstream steps.	High	Sprint-1

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
IT Procurement Team	Automated Workflow (Flow Designer)	USN-4	As a member of the IT procurement team, I want requests to automatically create tasks upon approval so that manual assignment overhead is eliminated.	The Flow Designer triggers automatically upon approval to generate the task without manual tracking	High	Sprint-2
Hardware Team Member	Task Assignment & Configuration	USN-5	As a hardware technician, I want catalog tasks automatically assigned to the "Hardware" group with a clear description so that I can configure laptops promptly	I can open the catalog task record, see the short description <i>“Laptop need to Configured”</i> , and find it assigned to the Hardware group	High	Sprint-2

ServiceNow Flow Designer Execution & Catalog Task Screenshot

Flow Designer: "New Laptop Request" (Active)

Flow Name: New Laptop Request | Trigger: Service Catalog | Status: Active

- Step 1: Catalog Item Requested (Completed)
- Step 2: Get Catalog Variables (Completed)
Variables: laptop_model, delivery_location, requested_by
- Step 3: Create Catalog Task (Running)
 - Action: Create Catalog Task
 - Catalog Item: Standard Laptop
 - Task Type: Fulfillment
 - Assignment Group: Hardware
 - Assigned To: — None —
 - Short Description: Fulfill Laptop Request for {requested_by}
 - Priority: Moderate
 - Due Date:
 - Description:

Generated Catalog Task: "SCTASK0045678"

ServiceNow Task View for SCTASK0045678: Fulfill Laptop Request

State: Open | Priority: 3 - Moderate | Assigned to: — None —

Number	Table Eds	Notes	Task
1	Assignment Group	SCTASK0045678	Priority: 3 - Moderate
2	Assignment Group	Hardware	Data Mapping & Execution
Short Description: Fulfill Laptop Request for Alex Johnson			
Description: Request details: Laptop Model: Dell XPS 15, Delivery: Building 4, Floor 2			
Request Item: RITM0045678			
Configuration Item:			

Notes: Variable Editor

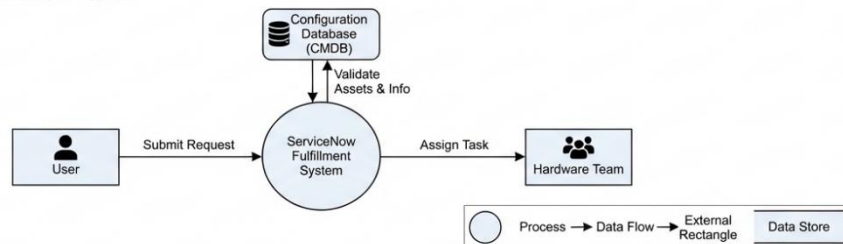
Assignment Group	SCTASK0045678	Value
Assignment Group	Hardware	Value

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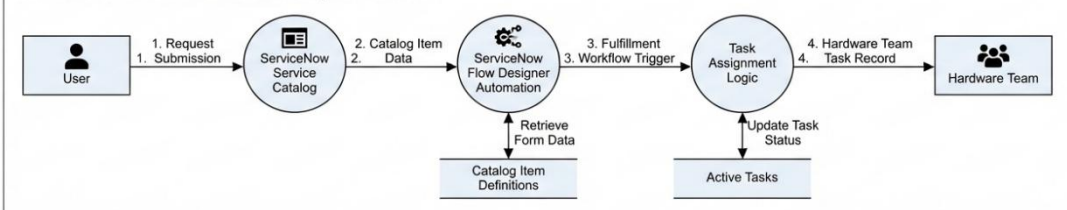
Data Flow Diagrams

DFD: Level 0 & Level 1 Diagrams

DFD Level 0: High-Level Fulfillment Context



DFD Level 1: Automated Hardware Request Process



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